

CYP3A4 Inhibition Prediction

What Is This Project?

CYP3A4 is an enzyme in your liver that breaks down drugs. If a new drug strongly inhibits this enzyme, it can cause serious drug-drug interactions one medication blocks the breakdown of another, causing overdose.

This project builds a machine learning model to predict, from molecular structure alone, whether a compound will inhibit CYP3A4. The model was trained on 5,506 known compounds (20% inhibitors, 80% non-inhibitors) and tested on 689 held-out compounds.

The Build

Two feature representations of molecules were used:

1. Morgan fingerprints: A 2D molecular fingerprint that captures local chemical patterns (what functional groups and bonds are present).
2. RDKit descriptors: Calculated molecular properties (weight, size, polarity, complexity, etc.). For CYP3A4, molecular weight is strong predictor heavier molecules tend to inhibit more.

Each feature type was fed into a Random Forest. The two predictions were then combined using logistic regression (meta-learner), creating an ensemble like asking two experts and averaging their opinions.

Results

Performance Metrics:

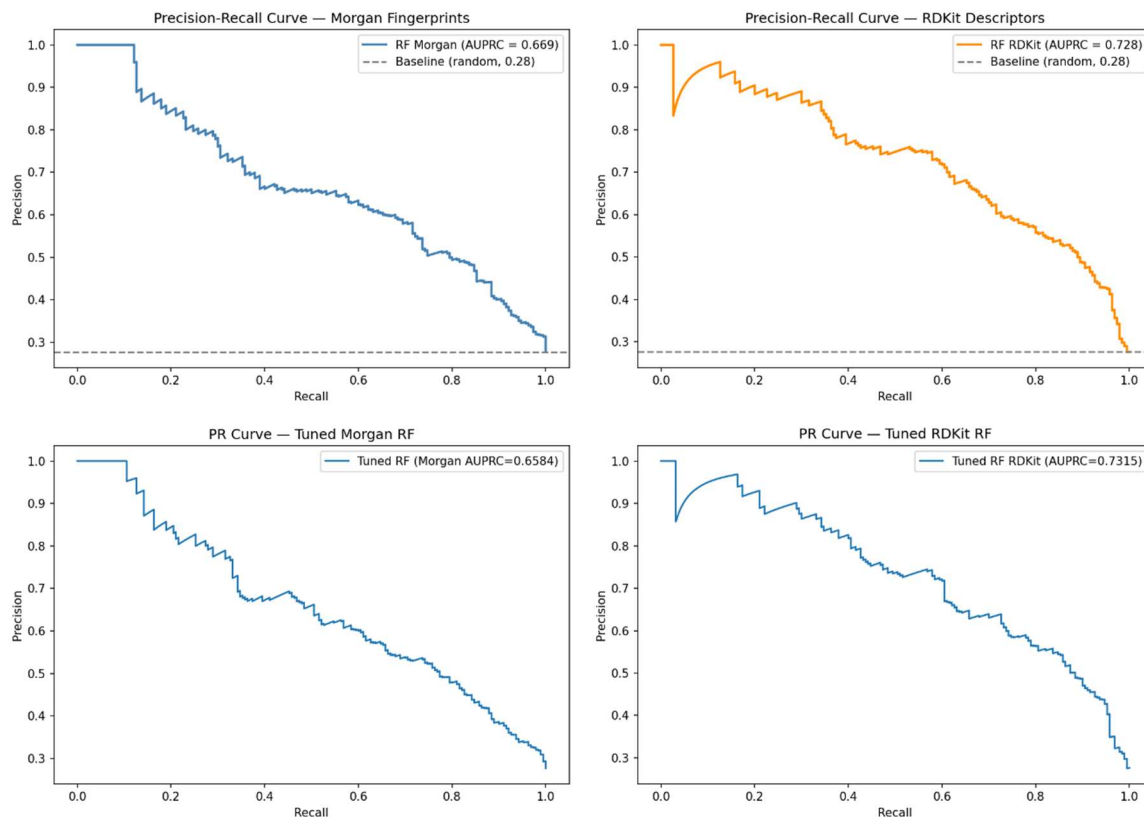
Model	Dataset	AUPRC
Morgan Fingerprints (RF alone)	Validation	0.669
RDKit Descriptors (RF alone)	Validation	0.728
Morgan + RDKit (Ensemble)	Validation	0.751
Ensemble (Final Test)	Test Set	0.710

What is AUPRC? Precision-Recall (PR) curves measure how well the model identifies the 20% of compounds that truly inhibit CYP3A4, while avoiding false alarms. AUPRC = Area Under the PR Curve. Higher is better; 0.20 = random guessing; 1.0 = perfect. The ensemble achieves 0.71, which is strong for this imbalanced dataset.

Key insight: Combining Morgan + RDKit improved performance by 4.5% over the best single model (RDKit alone at 0.728). This ensemble strategy works because the two feature types capture different information fingerprints encode structure, descriptors encode properties.

Precision-Recall Curves

Below are the PR curves showing model performance. Each curve plots how the model trades off precision (when it says 'inhibitor,' how often is it right?) against recall (of all true inhibitors, how many did we find?).



The dashed line at the bottom represents random guessing (AUPRC = 0.20). All four curves rise well above it, confirming the models learn real patterns. RDKit alone (orange) outperforms Morgan (blue), but combining them yields the best performance.

Structure-Activity Relationship Analysis

Structure-Activity Relationship (SAR) analysis answers: Which molecular scaffolds (core chemical frameworks) are more likely to inhibit CYP3A4?

What Are Scaffolds?

A scaffold is the core ring system of a molecule the basic skeleton without side chains. For example, benzene is a six-membered ring; benzyl is benzene with a methylene linker. Different scaffolds represent different chemical families, and compounds within the same family often share inhibition behaviour.

Key Findings

Scaffold Name	# Compounds	% Inhibitors	vs Baseline
Benzyl	71	49%	2.4× ↑
Aniline	28	43%	2.1× ↑
Benzene (baseline)	1,341	17%	—
Cyclohexane	61	10%	0.5× ↓
Furan	25	8%	0.4× ↓

What Does This Mean?

Enriched (High Inhibition): Benzyl and aniline scaffolds are 2 - 2.4 times more likely to inhibit CYP3A4 than random. These scaffolds feature:

- Larger molecular size (heavier compounds)
- Aromatic rings (extended π -electron systems that bind strongly to enzymes)

Depleted (Low Inhibition): Cyclohexane and furan are poor inhibitors. Why? They are:

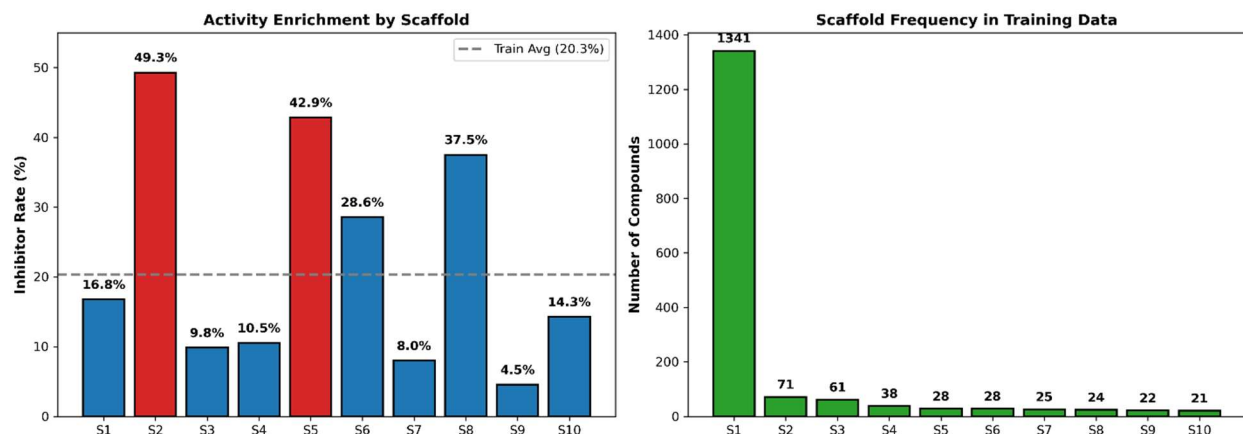
- Small and aliphatic (saturated, no aromatic character)
- Lack the size/complexity to fill CYP3A4's binding pocket

Bottom line: CYP3A4 preferentially binds large, aromatic molecules.

If you want to avoid CYP3A4 inhibition (good for side effects), use small aliphatic scaffolds.

If you want to inhibit CYP3A4 (e.g., for cancer treatment), use large benzyl or aniline cores.

SAR Visualization



Left panel: Activity rate by scaffold. Red bars = enriched (>40% inhibitors); blue = depleted (<40%). Right panel: How common each scaffold is in the training set.

Why Does This Matter?

The SAR analysis validates that the machine learning model learns chemically meaningful patterns. The model doesn't just memorize numbers it captures real structure-activity relationships known to medicinal chemists. This builds confidence that predictions on new compounds will be reliable.

For drug developers: These insights guide compound optimization. If a lead molecule has a cyclohexane core and you want to avoid CYP3A4 inhibition, keep it aliphatic. If the molecule has a benzyl core and you're worried about inhibition, you can predict it will likely be problematic without running an expensive lab assay.